Palash Yuvraj Ingle, Ph.D.

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Education

2020 – 2023 **Ph.D., Double Major, Security Engineering Lab.** Sejong University, South Korea.
Dept. (*Computer and Information Security, Convergence Engineering for Intelligent Drone)*
Thesis title: *Abnormal Object Video Synopsis in Multiview Cameras.*

2021 – 2022 **P.G.D in Artificial Intelligence and Machine Learning,** NIT Warangal (Online), India.

2017 – 2019 **M.Tech. Computer Science and engineering,** YMCA University, India.
Thesis title: *Using Hand Gesture for Solving Mute Deaf Problem and Control 3D Printed Prosthetic Hand Model Using IoT.*

2014 – 2016 **M.Sc. Information Technology,** Mumbai University, India.
Thesis title: *UGCV-Unmanned Ground Counter-IED Vehicle.*

2011 – 2014 **B.Sc. Information Technology,** Mumbai University, India.
Project title: *Live online examination system.*

Employment History

2023 – till date **Postdoctoral Researcher,** Security Engineering Lab, Sejong University, Seoul, South Korea.

2020 – 2023 **Research Scholar,** Security Engineering Lab, Sejong University, Seoul, South Korea.

2017 – 2020 **Assistant Professor (fixed)** Information Technology Department, K.J Somaiya College, mumbai, India.

Research Interests

- **Unmanned Aerial Vehicles**, Sensor Fusion, Object tracking and segmentation, LiDAR and Robotics.
- **Autonomous driving vehicle,** Three-dimensional object detection in the point cloud, Panoramic view of multiple cameras.
- **Video Synopsis**, Detection and extraction of abnormal objects in time and domain space, Stitching of foreground to background.

Projects

Ph.D 2020 – 2023

Development of artificial intelligence based video security technology and systems for public infrastructure safety:
The objective is to develop a video synopsis framework for detecting abnormal objects.
Technology: Python, Deep learning frameworks and library, Qt and Robot operating system, LiDAR, Drones, Depth Cameras.

Multiview abnormal video synopsis in real-time.
Objective:
Construction of a multiview abnormal video synopsis framework that in real-time detects and extracts abnormal objects from multiple videos to create a smaller video for analysis.
Funded by: Institute of Information and Communications Technology Planning and Evaluation (IITP) grant funded by the Korea government (MSIT) (No.2019-0-00231)
Technology: Python, C++ and MATLAB, Framework: TensorFlow and Keras, OpenCV, Hardware: IP-Cameras.

Dvs: A drone video synopsis towards storing and analyzing drone surveillance data in smart cities.
Objective:
Synchronization of multiple drones on the fly to conduct aerial surveillance for detecting and tracking abnormal objects by fusing LiDAR and camera data. Sensory UAV setup design, acquisition of point cloud data using LiDAR mounted on UAV using an onboard computer.
Funded by: Institute of Information and Communications Technology Planning and Evaluation (IITP) grant funded by the Korea government (MSIT) (No.2019-0-00231)
Technology: Python, C++, Framework: TensorFlow and Keras, Pytorch, OpenCV, ROS, Hardware: Depth AI Camera, gimbal camera, Velodyne Puck LITE, Tello, Parrot, DJI matrice.

Real-time abnormal object detection for video surveillance in smart cities.
Objective:
Development of the object detection model which detects smaller abnormal objects and performs better than the state-of-the-art detection model.
Funded by: Institute of Information and Communications Technology Planning and Evaluation (IITP) grant funded by the Korea government (MSIT) (No.2019-0-00231)
Technology: Python, C++, Framework: TensorFlow and Keras, OpenCV.

2017 – 2020

Development of Hand Gesture Model for mute deaf and construction of a 3D printed prosthetic hand model.
The objective is to develop a sensory-based hand gesture model that sends an instruction to an application that will speak. Construction of a 3D-printed hand to depict the hand gesture based on an impulsive sensor.
Technology: Python, C++, Deep learning frameworks and library, MIT App Inventor, 3D printer, Flex sensor, Arduino Uno, Raspberry pi.

Projects (continued)

2014 – 2017

Development of prototype: Unmanned Ground Counter-IED Vehicles (UGCV)
The objective is to create a prototype that will autonomously detect and defuse the IED explosive, the UGCV can work autonomously and can be controlled remotely. A 360-degree movement robotic arm was constructed to suit the scenarios which was mounted on a driving vehicle.
Technology: Confidential.

Research Publications

Journal Articles

1. **P. Y. Ingle** and Y.-G. Kim, "Multiview abnormal video synopsis in real-time," *Engineering Applications of Artificial Intelligence, SCIE, Impact Factor in JCR 2023: 8 JCR Ranking: 4.89%*, vol. 123, p. 106 406, 2023. URL: https://doi.org/10.1016/j.engappai.2023.106406.
2. **P. Y. Ingle** and Y.-G. Kim, "Video synopsis algorithms and framework: A survey and comparative evaluation," *Systems , SSCI, Impact Factor in JCR 2021: 2.895 JCR Ranking: 28.12%*, vol. 11, no. 2, p. 108, 2023. URL: https://doi.org/10.3390/systems11020108.
3. M. Parab, A. Bhanushali, **P. Ingle**, and B. Pavan Kumar, "Image enhancement and exposure correction using convolutional neural network," *SN Computer Science, Scopus, Impact Factor in JCR 2023: 3.78 JCR Ranking: 35%*, vol. 4, no. 2, p. 204, 2023. URL: https://doi.org/10.1007/s42979-022-01608-w.
4. **P. Y. Ingle** and Y.-G. Kim, "Real-time abnormal object detection for video surveillance in smart cities," *Sensors, SCIE, Impact Factor in JCR 2023: 3.576 JCR Ranking: 21.09%*, vol. 22, no. 10, p. 3862, 2022. URL: https://doi.org/10.3390/s22103862.
5. **P. Y. Ingle**, Y. Kim, and Y.-G. Kim, "Dvs: A drone video synopsis towards storing and analyzing drone surveillance data in smart cities," *Systems, SSCI, Impact Factor in JCR 2023: 2.895 JCR Ranking: 28.12%*, vol. 10, no. 5, p. 170, 2022. URL: https://doi.org/10.3390/systems10050170.

Patent

1. **P. Y. Ingle** and Y.-G. Kim, "Method and system for generating video synopsis based on abnormal object detection," in *Korea Institute of patent Information*, kipris, 2021.

Seminars and Talks

- Influence of artificial intelligence technology in human society and why it is not a threat.
- Real and fake research and its long-term impact on the Economy.
- The inclusion of CNN in video surveillance and its corresponding applications.
- Threats of using the internet and measures to safeguard children.
- Every human has a pattern and how significant it is in designing an artificial brain.
- How autonomous vehicles and drone technology can drive a country's economy in the next decades.
- Who is watching you: The danger behind using mobile phones and the internet and how everyone is being flawed.
- How to accommodate artificial intelligence technology in designing smart cities.

Skills

Coding	Python (Preference), Java, C, C++, C#, Matlab, HTML, Swift, Quantum computing language (QCL), JavaScript.
High Level Library	TensorFlow, PyTorch, Depth AI, OpenCV, DroneKit, Gazebo.
Sensor and Devices	IoT - NodeMCU, Arduino Uno, Raspberry Pi, Nvidia Jestson, Drones, LiDAR, etc.

Miscellaneous Experience

Research Awards and Achievements

2023 **Outstanding Research Award**, During Ph.D. at Sejong University.

2022 **Best Paper Award**, By Korea Information Processing Society (KIPS).

Bronze Best Paper Award, IEEE Seoul Section.

Bronze Paper Award, Korea Information Processing Society (KIPS-AOH).

Technical Certification

2009 **MS- CIT**. Awarded by Anubhav computer Insitute.

2012 **Certification in Computer Hardware and Networking**.Awarded by IDEMI (Gov of India).

2013 **Certification in C language**. Awarded by C-DAC, GIST.

Certification in C, C++, HTML, Core JAVA programming. By IDEMI (Gov of India).

Certification in C.Net and ASP.Net. Awarded by IDEMI (Gov of India).

2014 **Certification in Advance Animation and Film Making**. Awarded by IDEMI (Gov of India).

2016 **Master in Python Programming**. Awarded by Mumbai University.

2019 **Certified Ethical Hacker** . Awarded by RST.

CCNA. Awarded by RST.

Additional Certification

2018 **UGC NET**. Awarded by UGC (Gov. of India)

2012 **'C' Certificate Holder**. Awarded by NCC (Gov. of India).

2011 **Civil Defence Course.** Gov. of India.

Personal Information

Nationality- Indian.

Languages - English, Hindi, Marathi, level 1 Mandarin, and level 1 Korean.